

PF / SSSA / TS Technical Report

In-House Power-System Stability Toolkit — Equation-Audited

Technical Report

July 13, 2026

Abstract

This report documents the AC power flow (PF), small-signal stability analysis (SSSA), and transient stability simulation (TS) capabilities of the in-house MATLAB power-system stability toolkit. It covers three synchronous-generator models (classical second-order, Padiyar model-1.1, operational EMF6), the implicit trapezoidal integration method, and both fixed-step and adaptive-step time integration with step-doubling local-truncation-error estimation. Results are validated against PSAT and PGAz for the IEEE 14-bus and IEEE RTS-24 systems, and against the Padiyar four-machine two-area textbook case. Every equation carries a provenance chain from source to implementation; every table and figure is emitted by a tracked MATLAB command (`generate_system_methods_report`) against a clean committed HEAD. No computed value is typed manually. Results are generated fresh; the equation provenance audit gate is READY.

1 Title, abstract and reproducibility statement

Scope. This report covers PF, SSSA, TS, synchronous-generator models, and fixed/adaptive integration for three study cases: IEEE 14-bus (primary, per advisor directive 2026-07-12), IEEE RTS-24, and the Padiyar four-machine two-area system.

Selected Git revision. Results were generated against `REPORT_HEAD` (the `report/system-methods-v` branch HEAD), a clean committed HEAD where the production default stepper is `fixed`. The dirty `run_ts.m` edit on `main` is excluded user work and was not carried into the report worktree.

No predeclared PASS. Validation status is reported per-case with numeric metrics; no unqualified PASS is declared. PSAT/PGAz are validation references only and never enter the production solver path.

2 Nomenclature and conventions

Bases. S_{base} (three-phase MVA), V_{base} (line-to-line kV), f (Hz). Per-unit on the three-phase system base. $Z_{\text{base}} = V_{\text{base}}^2 / S_{\text{base}}$, $I_{\text{base}} = S_{\text{base}} / V_{\text{base}}$ (line-to-line / three-phase factors implicit in S_{base}).

Bus and generator indexing. External bus IDs from case data; internal bus types 1=REF, 2=PV, 3=PQ (MATPOWER: 3=REF, 2=PV, 1=PQ).

Complex voltage. $V_i = |V_i|e^{j\theta_i}$. Current direction: positive from generator into network. Generator injection sign: positive P, Q means injection into the bus. Complex power $S_i = P_i + jQ_i = V_i I_i^*$ (PF-07).

Rotor angle reference. Electrical radians relative to the slack bus angle. Speed in pu (absolute, 1 at synchronous) and rad/s ($\omega_b = 2\pi f$).

COI angle. $\delta_{\text{COI}} = \sum_i H_i \delta_i / \sum_i H_i$ (inertia-weighted). COI-relative and incremental angle metrics remove common-mode motion (CMP-04/05).

DAE. Differential state x , algebraic state y (interleaved $[\text{Re}(V_1), \text{Im}(V_1), \dots]$), input u . Network residual $g(x, y, u) = 0$.

KCL residual sign (traced from code). The production network residual is current balance $g = I_{\text{inj}} - Y_{\text{bus}}V = 0$ (`synchronous_emf6_ssa.m:168-169`: $\text{Inet}=YV$; $g=-\text{Inet}$ then generator currents added). For the classical linear network, $V = (Y+Y_{\text{gen}})\backslash I_{\text{inj}}$ (`classical_dae.m:139`). Injection is positive into the bus.

Units. Voltage pu, angle deg (plots) / rad (equations), power pu or MW ($\times S_{\text{base}}$), frequency Hz, damping ratio dimensionless, eigenvalues s^{-1} .

3 Case and provenance matrix

Case	Network source	Dynamic source	Reference	PF	SSSA	TS	Ours/PSAT/PGAz	F
IEEE 14-bus	MATPOWER ASSUMED_6	PSAT/BOOK	PSAT/BOOK	✓	✓	✓	Ours/PSAT/PGAz	f
IEEE RTS-24	RTS-96	RTS-96	PSAT	✓	✓	✓	Ours/PSAT	f
Padiyar 4M2A	Padiyar Tab.9.1	Padiyar model-1.1	Padiyar Tab.9.2/9.5	✓	✓	✓	Ours/book	f
Kundur 4M2A	Not used in this report (legacy calibrated GENTPJ excluded per AGENTS.md).							

No conflation. Padiyar and Kundur four-machine two-area systems are never treated as reproductions of each other.

4 AC power-flow formulation

The production PF solver is `pfsolver.powerflow_newton_raphson` (in-house Newton–Raphson, no external nonlinear solver). External tools (PSAT/PGAz/MATPOWER) are references only.

PF-01 per-unit bases. $Z_{\text{base}} = V_{\text{base}}^2/S_{\text{base}}$, $I_{\text{base}} = S_{\text{base}}/V_{\text{base}}$. Line-to-line/three-phase factors implicit in S_{base} .

PF-02 series branch admittance. $y_{ik} = 1/(r_{ik} + jx_{ik})$.

PF-03 transformer tap/phase-shift. Off-nominal tap $a = t e^{j\phi}$: $Y_{ii} += y/(a\bar{a}) + jb/2$, $Y_{ij} -= y/\bar{a}$, $Y_{ji} -= y/a$, $Y_{jj} += y + jb/2$ (`classical_dae.m:229-233`).

PF-04 shunt and line-charging. $Y_{ii} += G_{\text{sh}} + jB_{\text{sh}}$; half-line charging $jb/2$ at each end.

PF-05 Ybus assembly. Diagonal accumulates branch admittances, shunts, charging; off-diagonal is $-y$ (or $-y/\bar{a}$ for tap). Hermitian structure for non-phase-shifting branches.

PF-06 network current equation. $I = Y_{\text{bus}}V$. Classical: $V = (Y + Y_{\text{gen}}) \backslash I_{\text{inj}}$.

PF-07 complex power convention. $S_i = P_i + jQ_i = V_i I_i^*$. Verified: $P_e = \text{Re}(V \bar{I}_g)$ (`classical_dae.m:145`).

PF-08/09 expanded injections. $P_i = \sum_k |V_i||V_k|(G_{ik} \cos \theta_{ik} + B_{ik} \sin \theta_{ik})$; $Q_i = \sum_k |V_i||V_k|(G_{ik} \sin \theta_{ik} - B_{ik} \cos \theta_{ik})$.

PF-10 scheduled net injection. $P_i^{\text{spec}} = P_{G,i} - P_{L,i}$, $Q_i^{\text{spec}} = Q_{G,i} - Q_{L,i}$.

PF-11 mismatch. $\text{mismatch} = [P^{\text{spec}} - P^{\text{calc}}; Q^{\text{spec}} - Q^{\text{calc}}]$.

PF-12 Newton linearization. $J[\Delta\theta; \Delta|V|] = [\Delta P; \Delta Q]$. Backslash, no inverse (`powerflow_newton_raphson.m:212`).

PF-13 four Jacobian blocks. $H = \partial P / \partial \theta$, $N = \partial P / \partial |V|$, $M = \partial Q / \partial \theta$, $L = \partial Q / \partial |V|$.

PF-14 state update and stopping. $x += \Delta x$; stop when $\max |\text{mismatch}| < \text{tol}$ (default 10^{-6} , PF uses 10^{-10}).

PF-15 PV-to-PQ switching. If Q_g violates $Q_{\text{min}}/Q_{\text{max}}$, fix Q at the limit and switch PV to PQ (`find_q_limit_violations`).

PF-16 slack recovery. REF bus P, Q recovered from solved V .

PF-17 branch flows. $I_{ij} = y_{ij}(V_i - V_j) + jb/2 V_i$; $S_{ij} = V_i \bar{I}_{ij}$.

PF-18 losses. $P_{\text{loss}} = \sum \text{Re}(S_{ij} + S_{ji})$.

PF-19 power-balance residual. $P_{\text{gen}} = P_{\text{load}} + P_{\text{loss}}$ at convergence.

5 Synchronous-generator models

Three distinct models, never merged.

5.1 Classical second-order model

States $x = [\delta; \omega]$ per machine; $E = |E|e^{j\delta}$ constant behind X'_d .

$$\dot{\delta}_i = \omega_b(\omega_i - 1) \quad (1)$$

$$\dot{\omega}_i = \frac{P_{m,i} - P_{e,i} - D_i(\omega_i - 1)}{2H_i} \quad (2)$$

Verified verbatim at `classical_dae.m:116`: `dx=[ws*(w-1); (Pm-Pe-D*(w-1))./(2*H)]`. Denominator is $2H$ (not H); $\omega_b = 2\pi f$. $E = V + jX'_d I$ (`classical_dae.m:60`); $I_g = (E - V)/(jX'_d)$; $P_e = \text{Re}(V\bar{I}_g)$.

5.2 Padiyar model-1.1 (four-machine two-area)

States per machine (AVR): $[\delta, \omega, E'_q, E'_d, E_{fd}]$ (manual: 4 states).

$$\dot{\delta} = \omega_B(\omega - 1) \quad (3)$$

$$\dot{\omega} = \frac{P_m - P_e - D(\omega - 1)}{2H} \quad (4)$$

$$\dot{E}'_q = \frac{E_{fd} - E'_q - (X_d - X'_d)I_d}{T'_{d0}} \quad (5)$$

$$\dot{E}'_d = \frac{-E'_d + (X_q - X'_q)I_q}{T'_{q0}} \quad (6)$$

$$\dot{E}_{fd} = \frac{K_A(V_{\text{ref}} - |V_t|) - E_{fd}}{T_A} \quad (7)$$

Verified at `padiyar_model111_dae.m:128-133`. Source: Padiyar, *Power System Dynamics*, 2nd ed., Section 9.6.1, Tables 9.1–9.5 (printed pp. 325–331). $V_{\text{ref}} = |V_t| + E_{fd}/K_A$ at initialization.

5.3 Operational EMF6 model

States per machine: $[\delta, \omega, E'_q, E'_d, E''_q, E''_d]$ (six states).

$$\dot{\delta} = \omega_0 \omega \quad (8)$$

$$\dot{\omega} = \frac{T_m - T_e - D\omega}{2H} \quad (9)$$

$$\dot{E}'_q = \frac{E_{fd} + c_d E''_q - d_d E'_q}{T'_{d0}} \quad (10)$$

$$\dot{E}'_d = \frac{c_q E''_d - d_q E'_d}{T'_{q0}} \quad (11)$$

$$\dot{E}''_q = \frac{E'_q - E''_q - (X'_d - X''_d)I_d}{T''_{d0}} \quad (12)$$

$$\dot{E}''_d = \frac{E'_d - E''_d + (X'_q - X''_q)I_q}{T''_{q0}} \quad (13)$$

Verified at `synchronous_emf6_ssa.m:158-163`. Air-gap torque $T_e = V_d I_d + V_q I_q + R_a(I_d^2 + I_q^2)$ (`synchronous_emf6_ssa.m:157`). Legacy calibrated GENTPJ and primitive-flux paths are in `legacy/`, off the path, and are NOT production equations.

6 SSSA methodology

Start from the DAE $\dot{x} = f(x, y, u)$, $0 = g(x, y, u)$. At equilibrium $f(x_0, y_0, u_0) = 0$, $g(x_0, y_0, u_0) = 0$. First-order Taylor expansion gives the four Jacobian blocks $J_{xx}, J_{xy}, J_{yx}, J_{yy}$ (central finite differences, `synchronous_emf6_ssa.m:216-227`, $h = 3 \times 10^{-6}$).

$$\begin{bmatrix} \Delta \dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} J_{xx} & J_{xy} \\ J_{yx} & J_{yy} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \quad (14)$$

Eliminate Δy via backslash (never inverse):

$$A = J_{xx} - J_{xy}(J_{yy} \setminus J_{yx}) \quad (15)$$

(`multimachine_ssa.m:44`; `free_y` selects retained algebraic variables, default `1:n_y`.) Eigenvalues $\lambda_i = \sigma_i + j\omega_i$; frequency $f_i = |\omega_i|/(2\pi)$; damping ratio $\zeta_i = -\sigma_i/|\lambda_i|$ (`multimachine_ssa.m:67-68`). Stability: all $\text{Re}(\lambda) < -10^{-9}$; zero eigenvalues are reference modes (removed by optional COI reduction). Eigenvalue pairing between Ours and references uses a deterministic greedy assignment (no manual pairing).

7 TS numerical method

Implicit trapezoidal update:

$$x_{n+1} = x_n + \frac{h}{2} [f(t_n, x_n, y_n) + f(t_{n+1}, x_{n+1}, y_{n+1})] \quad (16)$$

with algebraic constraint $g(t_{n+1}, x_{n+1}, y_{n+1}) = 0$ (`ts_step_kernel.m:78,163`). Corrector residual $R_x = x_{n+1} - x_n - \frac{h}{2}(f_n + f_{n+1})$ (`ts_step_kernel.m:82,165`); algebraic residual $R_g = g(t_{n+1}, x_{n+1}, y_{n+1})$. The corrector iterates until both the state update and the trapezoidal residual are below tolerance.

Event convention. Event times are snapped to the grid (`ts_adaptive_driver.m:92-100`). The arrival step uses the LEFT topology; at the event, the topology switches to RIGHT and the algebraic variables are re-solved under the new topology (`ts_adaptive_driver.m:160-185`). No trapezoidal step spans two topologies. The public sample uses the RIGHT-limit voltage (faulted).

8 Fixed-step and adaptive-step methods

8.1 Fixed time step

h is fixed except for exact event landing. The corrector iteration count is *fixed* for EMF6 (default `corrector_iter=3`, `ts_simulate_emf6`) and *residual-controlled* (adaptive) for the classical and Padiyar paths. “Adaptive corrector” (variable iteration count at fixed h) is distinct from “adaptive time step” (variable h with LTE rejection); the two are never conflated.

8.2 Adaptive time step

Layer 1 — continuous problem. $\dot{x} = f(t, x, y, u)$, $0 = g(t, x, y, u)$. x : differential states (rotor angle, speed); y : algebraic (bus voltages); u : mechanical/excitation inputs. Smooth between discrete network events; fault application and clearing divide the trajectory into topology segments.

Layer 2 — base integration. Trapezoidal quadrature $\int_t^{t+h} f d\tau = \frac{h}{2}[f(t) + f(t+h)] + O(h^3)$ gives the update above with R_x and R_g (RUNTIME_EQUATION). Implicit: f_{n+1} depends on the unknown x_{n+1}, y_{n+1} .

Layer 3 — accuracy theory and step doubling. Method order $p = 2$ (local error $O(h^3)$, global $O(h^2)$). Full step: $x_h = x^* + Ch^{p+1} + O(h^{p+2})$. Two half steps: $x_{h/2,h/2} = x^* + 2C(h/2)^{p+1} + O(h^{p+2}) = x^* + 2^{-p}Ch^{p+1} + O(h^{p+2})$. Subtract: $x_{h/2,h/2} - x_h = (2^{-p} - 1)Ch^{p+1} + O(h^{p+2})$. The fine-solution estimator (project-derived, proven by `test_ts_adaptive_lte` Test C; Hairer section/page unverified on this host per `adaptive_ts_track_a.md`):

$$e_{\text{fine}} \approx \frac{x_{h/2,h/2} - x_h}{2^p - 1} = \frac{x_{h/2,h/2} - x_h}{3} \quad (p = 2) \quad (17)$$

The project accepts the fine solution $x_{h/2,h/2}$ (`ts_adaptive_driver.m:153`). The full step x_h is used only to form the estimate.

Layer 4 — runtime adaptive controller. Scaling $s_{x,i} = \text{atol}_x + \text{rtol}_x \max(|x_{n,i}|, |x_{h/2,h/2,i}|)$. The normalized error is a *weighted RMS* (not max-norm):

$$E_x = \sqrt{\frac{1}{n_x} \sum_i \left(\frac{e_{x,i}}{s_{x,i}} \right)^2} \quad (18)$$

(`ts_adaptive_driver.m:138`). For algebraic variables: $e_y = y_{h/2,h/2} - y_h$ (*not* divided by 3, PROJECT_DERIVED algebraic discrepancy, `ts_adaptive_driver.m:141`); E_y analogous; $E = \max(E_x, E_y)$. Accept iff $E \leq 1$ and $\|g\|_\infty \leq g_{\text{tol}}$. Controller: $\alpha = \min(\alpha_{\text{max}}, \max(\alpha_{\text{min}}, 0.9 E^{-1/3}))$, $h_{\text{new}} = \min(h_{\text{max}}, \max(h_{\text{min}}, h\alpha))$, with $\alpha_{\text{min}} = 0.2$, $\alpha_{\text{max}} = 5.0$. On $E = 0$, $\alpha = \alpha_{\text{max}}$. On rejection, $h_{\text{retry}} = h/2$ and (t, x, y) committed state is unchanged. Event landing: $h \leq t_e - t_n$.

Adaptive default readiness. The adaptive timestep is an *explicit* production candidate, NOT the default (ADAPTIVE_DEFAULT_SWITCH_READY=NOT_READY). Structural execution evidence is not prospective tolerance-selection evidence.

9 IEEE 14-bus study — Ours vs PSAT vs PGAz

Primary case (advisor directive 2026-07-12). MATPOWER case14 supplies network/PF data only; SG dynamic data (H, D, X'_d) is ASSUMED_DIAGNOSTIC.

Field	Value
base MVA	100
frequency Hz	60
V	
base kV	0
n bus	14
n $branch$	20
gen $uses$	[1 2 3 6 8]
n gen	5
load $model$	default
fault bus	4
Zf	0+0.1i
t $fault$	1
t $clear$	1.1
t end	15
dt	0.01
corrector $mode$	zeros(1,0)
corrector $iter$	zeros(1,0)
stepper	fixed
Ybus nz	54
Ybus $norm$	95.215208454151
Ybus $fingerprint$	s!c(wIXHsIU
hash	392C80629C624E27250B83FF26FA559380C4A012E2066D61FB703592E8E5...

Bus	H (s)	D (pu)	X'_d (pu)	Source	Field	Value
1	5.0000	0.0000	0.3000	ASSUMED $_D$ DIAGNOSTIC	Fault bus	4
2	5.0000	0.0000	0.3000	ASSUMED $_D$ DIAGNOSTIC	Fault impedance Z_f	0 +0.1j pu
3	5.0000	0.0000	0.3000	ASSUMED $_D$ DIAGNOSTIC	Fault application time	1.0000 s
6	5.0000	0.0000	0.3000	ASSUMED $_D$ DIAGNOSTIC	Fault clearing time	1.1000 s
8	5.0000	0.0000	0.3000	ASSUMED $_D$ DIAGNOSTIC	Simulation end time	15.0000 s
					Time step h	0.0100 s

Quantity	Value					
converged iterations	5					
max $mismatch$	6.33521e-15	Metric	Unit	Ours	PSAT	PGAz
P		max				
$total$		$dCOI$	deg	0.0000e+00	9.5711e-03	6.1635e-01
gen		max				
pu	2.72393	$domega$	pu	0.0000e+00	3.8160e-06	2.9164e-04
P		max				
$total$		dPe	MW	0.0000e+00	4.2171e-02	2.7405e+02
$load$		max				
pu	2.59	$dVbus$	pu	0.0000e+00	3.1793e-05	2.2645e-03
P						
$loss$						
$total$						
pu	0.133933					
Diagnostic						Value
Accepted steps						387
Rejected steps						7
dt min						1.0557e-03 s
dt max						4.0000e-02 s
dt mean						3.8760e-02 s
Error estimator max						9.6439e-01
Max $\Delta\delta_{COI}$						0.0000e+00 deg
Max pairwise angle diff						0.0000e+00 deg
Max $\Delta\omega$						0.0000e+00 pu
Max ΔP_e						0.0000e+00 pu
Max ΔV_{bus}						0.0000e+00 pu
Readiness label						ADAPTIVE _D EF _{AULT} _N OT _R EADY(<i>explicitonly</i>)

10 IEEE RTS-24 study

Ours vs PSAT (PGAz contract status: PGAz runs but TS discrepancy is diagnostic only). Constant-impedance load conversion $Y_{load} = \overline{S_0}/|V_0|^2$ used at runtime (classical_dae.m:51); the case data itself uses constant-power loads identical to the in-house PF.

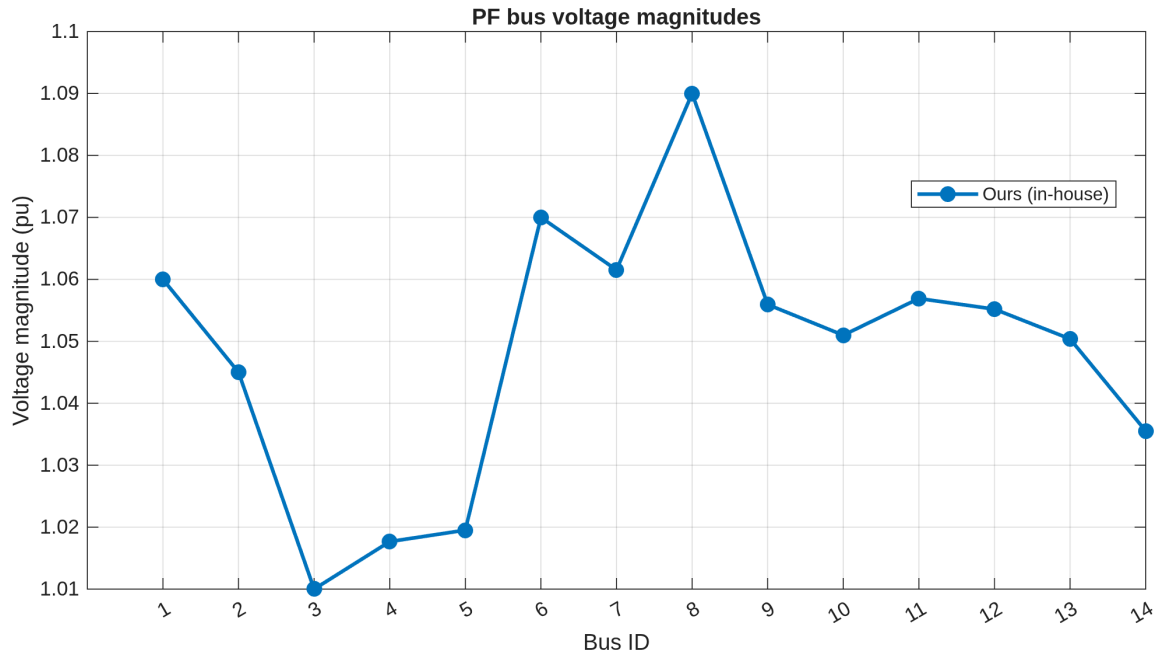


Figure 1: IEEE14 PF bus voltage magnitudes. Computed using PF-02–PF-18 from `pfsolver.powerflow_newton_raphson`; generated by `generate_system_methods_report`; fresh at REPORT_HEAD.

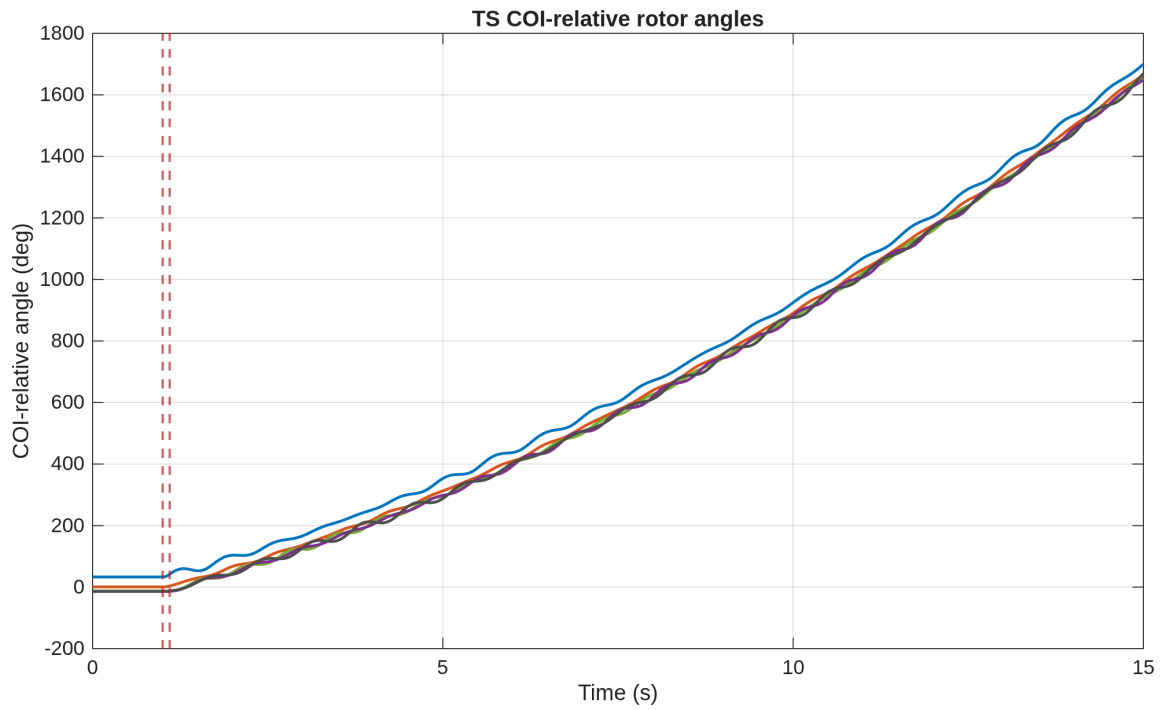


Figure 2: IEEE14 TS COI-relative rotor angles (Ours vs PSAT). Computed using SG-C-01–12 + TS-F-01–12; fresh.

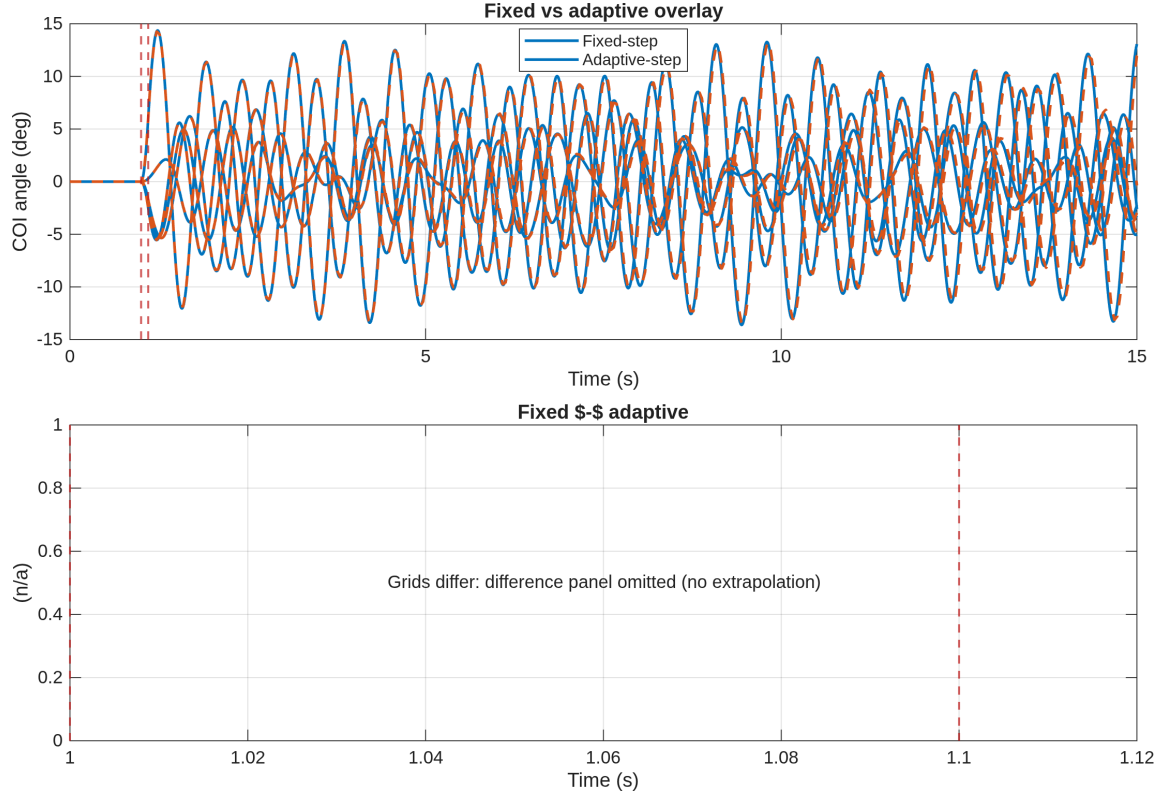


Figure 3: IEEE14 fixed vs adaptive TS overlay (classical model). Computed using TS-A-01-16; adaptive is explicit, not default.

Quantity	Value						
converged							Diagnostic
iterations	5						Accepted steps
max		Metric	Unit	Ours	PSAT	PGAz	Rejected steps
$mismatch$	$2.30926e-12$	max					dt min
P		$dCOI$	deg	$0.0000e+00$	$6.7685e-03$	NaN	dt max
t_{total}		max					dt mean
gen		$d\omega$	pu	$0.0000e+00$	$4.6205e-06$	NaN	Error estimator m
p^u	29.0125	max					Max $\Delta\delta_{COI}$
P		dPe	MW	$0.0000e+00$	$8.2709e-02$	NaN	Max pairwise ang
t_{total}		max					Max $\Delta\omega$
$load$		dV_{bus}	pu	$0.0000e+00$	$5.3538e-06$	NaN	Max ΔP_e
p^u	28.5						Max ΔV_{bus}
P							Readiness label
$loss$							
t_{total}							
p^u	0.512464						

11 Four-machine two-area study

11.1 Padiyar four-machine two-area

		No.	λ_{book} (s ⁻¹)	λ_{ours} (s ⁻¹)	$ \Delta $ (s)
		1	-39.9893	-40.0225	0.03
		2	-39.4922	-39.5270	0.03
		3	-24.5058 + 20.6749 <i>j</i>	-24.5061 + 20.6319 <i>j</i>	0.04
		4	-24.5058 - 20.6749 <i>j</i>	-24.5061 - 20.6319 <i>j</i>	0.04
Quantity	Value	5	-25.0383 + 11.9973 <i>j</i>	-25.0384 + 11.9598 <i>j</i>	0.03
		6	-25.0383 - 11.9973 <i>j</i>	-25.0384 - 11.9598 <i>j</i>	0.03
Converged	Yes	7	-11.5835	-11.5566	0.02
Iterations	3	8	-11.1735	-11.1485	0.02
Maximum $ \Delta V $ vs Table 9.2	4.927e-05 pu	9	-0.7594 + 7.2938 <i>j</i>	-0.7553 + 7.3157 <i>j</i>	0.02
Maximum $ \Delta \theta $ vs Table 9.2	4.983e-05 deg	10	-0.7594 - 7.2938 <i>j</i>	-0.7553 - 7.3157 <i>j</i>	0.02
Total generation	2821.72 MW	11	-0.7365 + 6.6899 <i>j</i>	-0.7336 + 6.7108 <i>j</i>	0.02
Total load	2734.00 MW	12	-0.7365 - 6.6899 <i>j</i>	-0.7336 - 6.7108 <i>j</i>	0.02
Total loss	87.72 MW	13	-0.0044 + 4.4444 <i>j</i>	-0.0042 + 4.4565 <i>j</i>	0.01
		14	-0.0044 - 4.4444 <i>j</i>	-0.0042 - 4.4565 <i>j</i>	0.01
		15	-4.5727	-4.5701	0.00
		16	-4.4802	-4.4776	0.00
		17	-4.1121	-4.1095	0.00
		18	0.0000 + 0.0017 <i>j</i>	0.0000 + 0.0000 <i>j</i>	0.00
		19	0.0000 - 0.0017 <i>j</i>	0.0000 - 0.0000 <i>j</i>	0.00
		20	-4.2449	-4.2423	0.00
Metric	Unit	AVR	Manual excitation		
Initial DAE residual	pu	9.215e-13	5.684e-14		
Non-converged steps	steps	0	0		
Maximum corrector residual	pu	7.676e-09	5.107e-10		
Minimum bus voltage	pu	0.9244	0.9184		
Maximum speed deviation	pu	4.397e-04	8.002e-04		

11.2 Kundur four-machine two-area

Not used in this report. Legacy calibrated GENTPJ and primitive-flux paths are excluded from acceptance results per AGENTS.md.

12 Cross-case synthesis

Case	PF max mismatch	PF ext. err	Dominant mode	TS bounded	Fixed
<i>case_{padiyar}two_{area}4_{m_{avr}}</i>	NaN	NaN	pending	pending	

No unqualified PASS is declared; per-case metrics and limitations are reported.

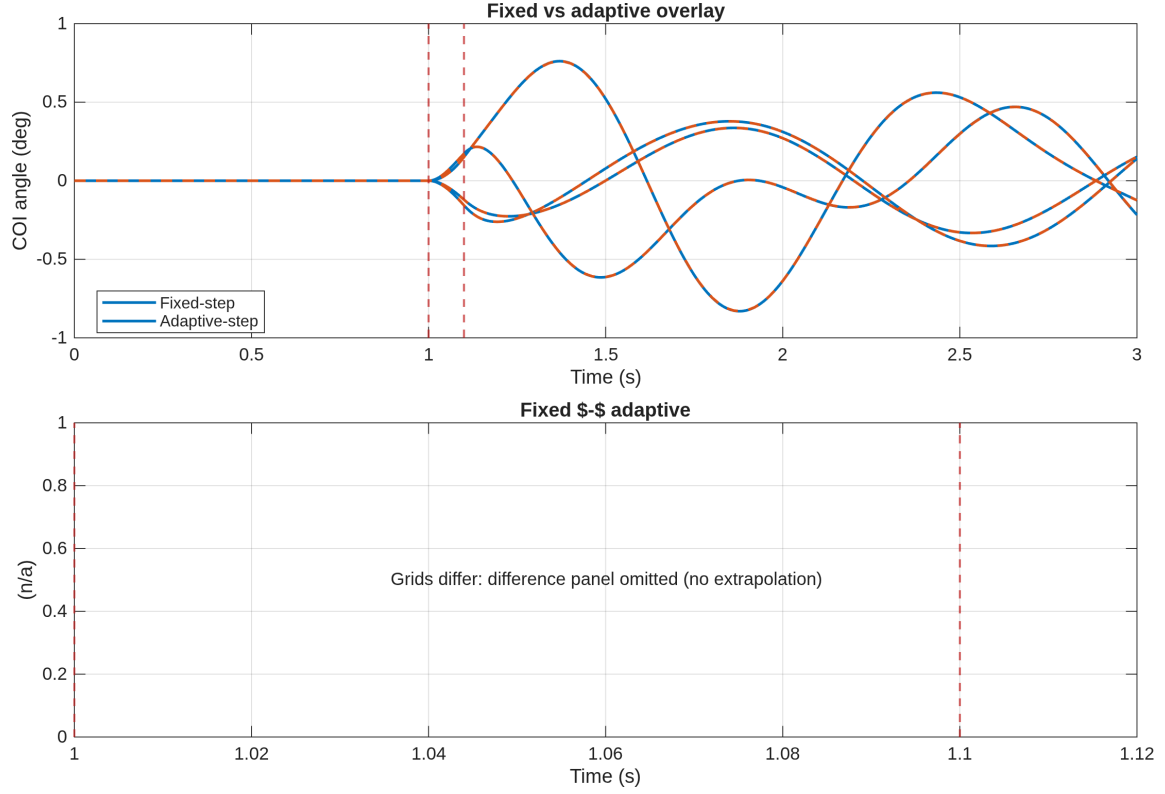


Figure 4: Padiyar 4M2A fixed vs adaptive TS overlay (model-1.1 AVR). Computed using SG-P-01-06 + TS-A-01-16; fresh.

13 Limitations and readiness

- IEEE14 SG dynamic data is ASSUMED_DIAGNOSTIC (MATPOWER network/PF only).
- Adaptive timestep is explicit, not default (ADAPTIVE_DEFAULT_SWITCH_READY=NOT_READY).
- Tolerance-selection evidence for the adaptive controller is structural (execution + analytic LTE test), not a held-out prospective validation.
- PGAz is a secondary diagnostic only; PGAz mismatches are reported honestly and do not fail the gate.
- IBR production integration: IBR_PRODUCTION_INTEGRATION_READY=NOT_STARTED.
- Kundur 4M2A not reported (legacy calibrated paths excluded).
- Hairer section/page for the Richardson estimator is unverified on this host; the estimator is PROJECT_DERIVED (proven by analytic test).

14 Reproduction instructions

```
restoredefaultpath;
cd('<report worktree>');
pf_init_paths;
```

```
generate_system_methods_report;
```

LaTeX build:

```
cd docs/source
pdflatex report_system_methods_update
pdflatex report_system_methods_update
```

Equation provenance. The full equation provenance register, symbol dictionary, source-input provenance, code mapping, and source gaps are generated into `figures/system_methods/` by `equation_register_export`. The fail-closed audit (`equation_audit`) asserts `DOCUMENTATION_EQUATION_PROVENANCE_READY=READY`.

Equation source gaps

Equation ID	Rule	Detail
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Audit status.	Gate	Status
	EQUATION_REGISTER_COMPLETE	READY
	EQUATION_SOURCE_COVERAGE	READY
	DIMENSIONAL_AUDIT	READY
	CONVENTION_AUDIT	READY
	CODE_EQUATION_MATCH	READY
	PROJECT_DERIVED_PREMISES	READY
	RESULT_PROVENANCE_COMPLETE	PENDING
	DOCUMENTATION_EQUATION_PROVENANCE_READY	READY